

FRTT350J Rear Helper Spring Kit Installation Manual

Model: FRTT350J (Rear Suspension Package)

Please read these instructions carefully before starting the installation. Ensure all standard safety precautions are observed while working underneath the vehicle.

1. Parts List

Before beginning, verify that all components listed below are present in your kit.

Ref #	Description / Specifications	Qty
1	Weldment Bracket (Left-Hand & Right-Hand)	2
2	Hollow Rubber Spring	2
3	Hex Cap Screw (M10 x 40 x 1.5, Grade 8.8, Zinc Plated)	2
4	Hex Cap Screw (M10 x 65 x 1.5, Grade 8.8, Zinc Plated)	2
5	Shim Plate (1-3/8" Length x 1/8" Width x 1/32" Thickness, 5/8" Slot)	4
6	Rubber Spacer (4" Outer Diameter x 1" Thickness, with Bushing)	2
7	Fender Washer (3/8" Inner Diameter x 1-1/2" Outer Diameter, Grade 2, Zinc Plated)	2

2. Installation Instructions

- 1. Prepare Vehicle and Remove OEM Bump Stop:** Safely elevate the vehicle using an appropriate lift or jack stands, ensuring it is stable and secure. Remove the OEM bump stop assembly, retaining any fasteners for potential re-use.
- 2. Assemble Helper Spring Unit:** Position the hollow rubber spring (Ref 2) on the weldment bracket (Ref 1). If the spacer is not used, secure the assembly with the shorter M10 x 40 hex cap screw (Ref 3) and fender washer (Ref 7).
- 3. Install Helper Spring Assembly:** Mount the assembled helper spring unit to the vehicle frame utilizing the existing OEM mounting holes. Use retained OEM fasteners and any necessary shim plates (Ref 5) for proper alignment and fitment. If an OEM peg-style casting is present, position the spring to contact the peg; do not remove the OEM peg.
- 4. Adjust Clearance (Optional):** If unloaded clearance is excessive, install the 1" rubber spacer (Ref 6) between the spring and bracket and secure it with the longer M10 x 65 hex cap screw (Ref 4) and fender washer (Ref 7). Do not use the spacer if it causes insufficient clearance or preload.
- 5. Final Inspection:** Verify that all fasteners are tightened securely to the vehicle manufacturer's recommended torque specifications. Ensure no brake lines, brake cables, or other vehicle components interfere with the helper spring assembly during suspension travel or under load.

Installation Notes:

- Verify that brake lines or cables do not make contact with the helper spring assembly during suspension travel.
- If the OEM vehicle is equipped with a peg-style casting at the bump stop location, the helper spring is designed to make contact with this peg. Do not remove the OEM peg.

3. Clearance Instructions & Fitting Guidance

- **Clearance Gap:** With the vehicle unloaded and at normal ride height, maintain approximately **1/2" to 1-1/2" (13-38 mm)** of clearance between the helper spring and its contact point. Approximately **1" (25 mm)** is a practical target.
- **Ride Comfort:** Proper clearance helps maintain a comfortable unloaded ride. Insufficient clearance or excessive preload may result in a firmer ride.
- **Spacer Use:** Use the included **1" spacer** only when the unloaded gap is excessive. If the clearance is already within the recommended range, the spacer is not required.
- **Load Effects:** A partial load or additional vehicle accessories may reduce or eliminate this initial clearance gap.

Important Safety Warnings:

- Do not exceed the vehicle manufacturer's recommended axle weight rating / GAWR.
- Inspect for potential interference between the hollow rubber spring and adjacent vehicle components. Under load, the rubber spring will expand in diameter.

Note: Illustrations are for reference only. Vehicle configuration and actual component appearance may vary. Install according to the vehicle-specific mounting points and standard mechanical practices.